

CBCL Severity Prediction from Parent Free-Text

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<https://github.com/OfriA/AppliedProject52056>

Abstract

This study examines whether parent-authored free text collected during wartime can be used to predict children’s behavioral and emotional difficulties as measured by the Child Behavior Checklist (CBCL). We formulate the problem as a three-class severity prediction task and compare a linear baseline based on multilingual text embeddings with a pretrained multilingual transformer, examining both textual information alone and text combined with child age.

The results indicate that parental texts contain predictive information about child difficulties, but that this relationship is challenging to model consistently. The strongest model performed better than the linear baseline, although its advantage was sensitive to the train-test split. Qualitative analysis further suggested that parental distress, coping language, and war-related context can complicate the relationship between the text and the child-level outcome. An additional ordinal-classification experiment did not improve prediction. Overall, the findings highlight both the potential and the limitations of using parent-authored text to infer children’s behavioral and emotional difficulties from a small and imbalanced dataset.

1 Introduction

Armed conflict can substantially affect children’s psychological well-being, and children’s adjustment is closely related to the family environment, including caregiver distress and parenting behavior (Keleynikov et al., 2025a). During the early stages of the Israel–Hamas war, parents faced the dual challenge of coping with their own distress while supporting their children through fear, uncertainty, and disrupted routines (Keleynikov et al., 2025b).

The data used in this project were collected as part of a broader study of parenting and emotion regulation during the war. Parents described an upsetting parenting-related event and wrote a self-directed emotion-regulation letter, while also re-

porting their child’s behavioral and emotional difficulties using the Child Behavior Checklist (CBCL) (Keleynikov et al., 2025b,a). We ask whether these parent-authored texts contain enough information to predict the severity of the child’s difficulties. We formulate the task as three-class classification (Low, Medium, and High CBCL severity), with particular interest in identifying High-severity cases. The task is challenging because the texts primarily describe the parent’s experience rather than the child’s condition, and because High-severity observations are relatively rare.

2 Exploratory Analysis

2.1 Dataset

The data originate from a broader study of parenting and emotion regulation during the Israel–Hamas war (Keleynikov et al., 2025b,a). The raw dataset contained 659 observations and 344 variables, consisting primarily of parent-reported questionnaire measures, demographic information, and two free-text fields. For the present prediction task, we retained observations with complete CBCL and war-exposure measurements and a non-missing emotion-regulation letter, resulting in a final sample of 270 observations. Each retained observation corresponds to a parent reporting about one child.

The two textual inputs were collected as part of an emotion-regulation task (Keleynikov et al., 2025b). In the *Event* field, parents briefly described a personally upsetting parenting-related event associated with the war. In the *EER_text* field, parents subsequently wrote a self-directed letter intended to help alleviate the negative emotions elicited by that event. The responses were predominantly written in Hebrew and were highly variable in both length and writing style, ranging from short fragments to detailed narratives and including informal language, spelling variation, and typographical errors.

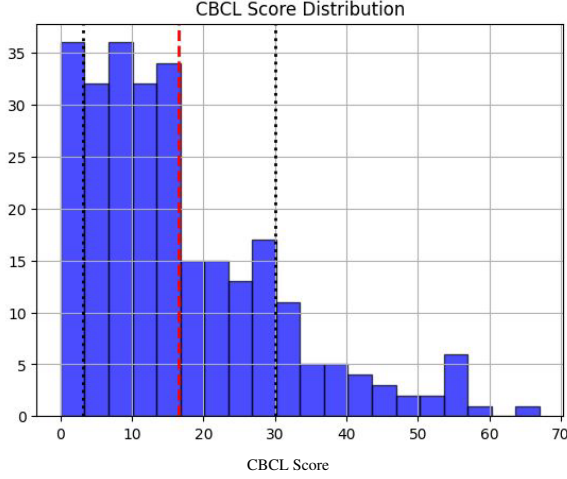


Figure 1: Distribution of total CBCL scores in the final analysis sample. Scores are right-skewed, with most observations concentrated at lower values and relatively few observations in the upper tail.

Variable	Mean	SD
CBCL total	16.62	13.91
Depression/anxiety	6.54	4.99
Aggressive behavior	7.54	7.40
Somatic complaints	2.55	3.63
Event length (characters)	124.44	101.91
EER_text length (characters)	85.34	113.42

Table 1: Descriptive statistics for the prediction target and textual inputs.

2.2 Prediction Target

Children’s behavioral and emotional difficulties were assessed using selected subscales from the Child Behavior Checklist (CBCL) (Achenbach, 1991). Following the original study from which the data were obtained, we used 43 items from three CBCL subscales: depression/anxiety, aggressive behavior, and somatic complaints (Keleynikov et al., 2025a).

The total CBCL score had a mean of 16.62 ($SD = 13.91$) and exhibited a pronounced right-skewed distribution (Figure 1). Most observations were concentrated at relatively low scores, whereas progressively fewer children appeared toward the upper tail of the distribution. This pattern is important for the prediction task because observations with more severe behavioral and emotional difficulties are comparatively scarce, making them harder to learn from and evaluate reliably. Additional descriptive statistics for the CBCL components and the two textual inputs are reported in Table 1.

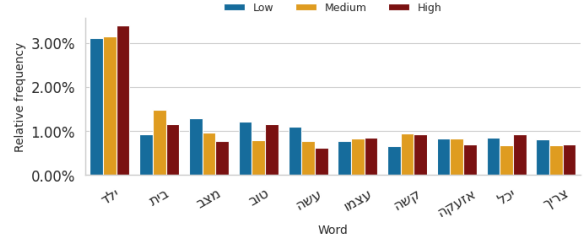


Figure 2: Normalized frequencies of the most common words across the Low, Medium, and High CBCL severity groups. The dominant vocabulary shows substantial overlap across groups, with no clear lexical pattern separating the High-severity observations.

2.3 Exploratory Text Analysis

Beyond their different lengths, the two text fields also contained qualitatively different types of information. Event descriptions generally focused on the stressful situation itself, including children’s fears, disrupted routines, exposure to war-related information, and parents’ feelings of helplessness. In contrast, the self-directed letters often contained reassurance, self-compassion, positive reframing, and attempts to reduce distress. This distinction suggested that the two fields could provide complementary information: the Event text describes the stressor and the parent’s immediate experience, whereas the EER text reflects the parent’s subsequent emotional processing.

We also examined normalized word frequencies across the CBCL severity groups. Figure 2 shows the most frequent words and their relative frequencies within each group.

Among the ten most frequent words overall, the dominant vocabulary showed substantial overlap across the Low, Medium, and High groups. Some words exhibited graded differences across severity levels, such as “child”, which increased from Low to High, while others showed the opposite pattern. However, these trends were not consistent across the vocabulary, and no clear lexical separation of the High-severity group emerged. This suggested that the relevant predictive signal was more likely to depend on contextual usage and combinations of words than on the frequency of individual common words, motivating the use of contextual multilingual text representations in Section 3.

3 Training and Evaluation Setup

3.1 Target Construction and Data Split

Although the original CBCL outcome is numerical, our primary objective was not to predict an exact score but to distinguish broad levels of behavioral and emotional difficulty. From a practical perspective, small differences in the predicted CBCL score are less informative than identifying children who fall toward the upper end of the difficulty distribution and may warrant greater attention. We therefore reformulated the task as a three-class classification problem, allowing the models to focus on broader severity groups rather than fine-grained numerical prediction. These groups were defined for the purposes of the present prediction task and should not be interpreted as clinical diagnostic or treatment thresholds.

The final dataset was randomly divided into 80% training data ($n = 216$) and 20% held-out test data ($n = 54$).

To avoid using information from the test set when defining the prediction target, the class boundaries were calculated exclusively from the CBCL distribution in the training set. The training-set median was 13, and the 85th percentile was 30. Observations with CBCL scores at or below 13 were labeled *Low*; observations above 13 and at or below 30 were labeled *Medium*; and observations above 30 were labeled *High*. Thus, the training labels contained approximately 50% Low, 35% Medium, and 15% High observations. The same numerical thresholds were then applied unchanged to the test set. This procedure preserved the held-out test set for estimating generalization performance while reflecting our particular interest in the upper tail of the CBCL distribution.

3.2 Text Representation

For the linear baseline, the two textual fields were encoded independently using the pretrained `intfloat/multilingual-e5-base` multilingual text-embedding model (Wang et al., 2024). Each text was represented by a normalized 768-dimensional vector. The representations of *Event* and *EER_text* were then concatenated, producing a 1,536-dimensional representation for each observation. This representation was used as the text-only feature set for the logistic-regression baseline. The alternative feature set, which additionally incorporates the child’s age, is described in Section 4.

3.3 Evaluation Metrics

The class imbalance and our emphasis on the relatively rare High-severity group motivated the use of two complementary evaluation metrics. We used macro-averaged F1 as the primary measure of overall multiclass performance. For each class c , the F1 score is the harmonic mean of precision and recall, and Macro-F1 is the unweighted average across the three classes:

$$F1 = \frac{2PR}{P + R}, \quad (1)$$

$$\text{Macro-F1} = \frac{F1_{Low} + F1_{Medium} + F1_{High}}{3}. \quad (2)$$

By averaging the class-specific F1 scores equally, Macro-F1 provides a balanced measure of overall model performance across Low, Medium, and High, regardless of their prevalence.

In addition, we report the F_2 score specifically for the High class. For High-class precision P and recall R ,

$$F_2 = \frac{5PR}{4P + R}. \quad (3)$$

Whereas Macro-F1 summarizes performance across all three severity groups, High-class F_2 focuses specifically on the model’s ability to identify the relatively rare High-severity observations. The F_2 score places greater emphasis on recall than precision, reflecting our preference for detecting a larger proportion of High cases even at the cost of additional false-positive predictions.

3.4 Experimental Protocol

The held-out test set was not used for model selection or hyperparameter tuning. For the logistic-regression baseline, hyperparameters were selected within the training data using five-fold stratified cross-validation, with the High-class F_2 score as the optimization criterion. For the transformer-based models, the training portion was further divided internally into training and validation subsets for model development, while the original held-out test observations remained unchanged. All primary model comparisons were performed on the same 80/20 split and using the same evaluation metrics.

Because the original test set contained only a small number of High-severity observations, class-specific evaluation metrics were particularly sensitive to individual predictions. In the primary 80/20

split, the test set contained only nine High-severity observations, such that correctly or incorrectly classifying a single case changed High-class recall by approximately 11 percentage points and could substantially affect the corresponding F_2 score. We therefore repeated the transformer evaluation experiment using 70/30 and 60/40 train–test allocations as a robustness analysis. These alternative splits were not used to replace the primary experimental setup, but rather to assess how sensitive the observed conclusions were to the amount of data allocated to training and testing. The results of this robustness analysis are reported in Section 5.2.

4 Prediction Models

We compare two model families under two feature settings. The first feature set contains only the two parental text fields, *Event* and *EER_text*. The second additionally incorporates the child’s age. This yields four primary configurations: logistic regression with text only, logistic regression with text and age, XLM-RoBERTa with text only, and XLM-RoBERTa with text and age. Keeping the feature comparison parallel across the two model families allows us to examine the contribution of child age within each modeling approach.

4.1 Multinomial Logistic Regression

Our baseline model was multinomial logistic regression applied to the concatenated multilingual E5 embeddings described in Section 3. This baseline tests whether the three severity groups can be separated by a linear decision boundary in the fixed embedding space. The inverse regularization parameter C , which controls the strength of regularization, was selected using five-fold stratified cross-validation on the training data, optimizing High-class F_2 . The selected value for the text-only model was $C = 3$.

For the second feature setting, child age was added to the 1,536-dimensional text representation. Age was standardized using the mean and standard deviation of the training data, and the same training-derived transformation was then applied to the test data, avoiding test-set leakage. The standardized age value was concatenated with the text embeddings. The same cross-validation procedure was repeated for this feature setting, resulting in a selected value of $C = 10$.

4.2 XLM-RoBERTa Classifier

To move beyond a linear classifier over fixed embeddings, we used xlm-roberta-base as the advanced multilingual text model (Conneau et al., 2020). We selected XLM-RoBERTa because its encoder architecture is well suited to text-classification tasks, while its multilingual pretraining makes it appropriate for our predominantly Hebrew data. This combination allows the model to capture contextual information in the parental texts that may be relevant for distinguishing between CBCL severity classes. Given the limited sample size, the pretrained transformer backbone was frozen and only the task-specific classification layers were trained, reducing the number of trainable parameters while retaining the contextual multilingual representations learned during pretraining.

Class imbalance was addressed using partial random oversampling within the inner training set only. In the primary experiment, the inner training distribution of 89 Low, 58 Medium, and 25 High observations was adjusted to 89, 70, and 50 observations, respectively. We deliberately avoided complete class balancing because this would require extensive duplication of the small number of High-severity observations. Validation and test observations were never oversampled.

The text-only and text-plus-age models were trained using the same configuration to ensure a controlled comparison between feature sets. In the text-only setting, the *Event* and *EER_text* fields were provided as a single structured textual input. In the second setting, child age was appended to the same sequence as textual metadata. Full input formatting and training hyperparameters are provided in Appendix A.

5 Results

5.1 Primary Model Comparison

Table 2 presents the primary comparison between the two model families and the two feature sets on the held-out test set. Adding child age did not provide a clear improvement to the logistic-regression baseline: overall performance remained similar, while the age-augmented model correctly identified one fewer High-severity observation. In contrast, adding child age to XLM-RoBERTa produced a substantially larger improvement on the primary split, increasing both Macro-F1 and the detection of High-severity observations.

Model	Features	Macro-F1	High Prec.	High Rec.	High F_2
Logistic Regression	Text	0.368	0.154	0.222	0.204
Logistic Regression	Text + age	0.347	0.091	0.111	0.106
XLM-RoBERTa	Text	0.428	0.125	0.111	0.114
XLM-RoBERTa	Text + age	0.518	0.385	0.556	0.510

Table 2: Performance of the two model families under the two feature settings on the primary 80/20 split. Bold values indicate the best result for each metric.

Using text alone, XLM-RoBERTa achieved higher overall multiclass performance than logistic regression, increasing Macro-F1 from 0.368 to 0.428. However, this improvement did not extend to the High class: High-class F_2 decreased from 0.204 to 0.114, with the XLM-RoBERTa text-only model correctly identifying only one of the nine High observations. Thus, the contextual language model improved overall classification performance but did not, by itself, improve detection of the rare High-severity group.

The effect of adding child age differed across the two model families. For logistic regression, adding age did not provide a clear improvement: Macro-F1 changed from 0.368 to 0.347, while the number of correctly identified High observations decreased from two to one. In contrast, adding child age to XLM-RoBERTa produced the strongest performance in the primary experiment. Macro-F1 increased from 0.428 to 0.518, High recall increased from 0.111 to 0.556, and High precision increased from 0.125 to 0.385, resulting in a High-class F_2 of 0.510.

XLM-RoBERTa with child age was therefore the best-performing configuration on the primary split. The contrasting results across model families indicate that adding age did not provide a uniform benefit across modeling approaches. Moreover, because the held-out test set contained only nine High observations, the magnitude of the improvement observed for XLM-RoBERTa should be interpreted cautiously and examined for robustness across alternative data splits.

5.2 Robustness to Train-Test Split

The strong performance of the XLM-RoBERTa model with child age on the primary 80/20 split was obtained from a relatively small test set. To examine whether this result was robust to the particular data allocation, we repeated the XLM-RoBERTa experiments using 70/30 and 60/40 train-test splits. The same modeling approach was evaluated both with text only and with child age added to the input.

The advantage of adding child age observed in

Split	Features	Macro-F1	High F_2
80/20	Text	0.428	0.114
80/20	Text + age	0.518	0.510
70/30	Text	0.390	0.290
70/30	Text + age	0.358	0.375
60/40	Text	0.368	0.147
60/40	Text + age	0.297	0.000

Table 3: XLM-RoBERTa performance across alternative train-test splits. Bold values indicate the better feature setting within each split and metric.

the primary split was not consistent across the alternative data allocations. Under the 70/30 split, adding age increased High-class F_2 from 0.290 to 0.375, while Macro-F1 was slightly lower, decreasing from 0.390 to 0.358. Under the 60/40 split, the text-only model retained a Macro-F1 of 0.368 and a non-zero High-class F_2 of 0.147. In contrast, adding child age reduced Macro-F1 to 0.297, and the model failed to identify any High observations, resulting in a High-class F_2 of 0.

These results indicate sensitivity to the train-test allocation. In particular, the large improvement associated with child age in the primary 80/20 experiment should be interpreted cautiously rather than as a stable effect. The comparison also highlights an important trade-off in a small dataset. Allocating more observations to training provides the model with more examples from which to learn, particularly from the rare High class, but leaves a smaller test set in which evaluation metrics are highly sensitive to individual predictions. Conversely, a larger test set provides a more informative basis for evaluating generalization, but reduces the amount of data available for model training.

6 Model Analysis

We analyzed the behavior of the logistic-regression baseline and the strongest advanced model, XLM-RoBERTa with child age, using three complementary approaches. First, a permutation-based null-distribution analysis examined whether the observed predictive performance exceeded what could be obtained after breaking the relationship

between the inputs and the target. Second, a quantitative error analysis examined class-specific performance, with particular attention to the rare High-severity class. Finally, a qualitative analysis examined representative predictions in order to identify recurring textual patterns associated with model successes and failures.

6.1 Null-Distribution Analysis

To assess whether model performance reflected a meaningful relationship between the inputs and the CBCL outcome, we constructed permutation-based null distributions. For each model, the training labels were randomly permuted 100 times, the model was retrained using the same pipeline, and performance was evaluated on the unchanged held-out test set. Figure 3 compares the observed Macro-F1 and High-class F_2 with their corresponding null distributions.

For the logistic-regression baseline, the observed Macro-F1 was approximately at the 75th percentile of the null distribution, suggesting that the model captured some overall predictive signal beyond what was typically obtained after breaking the relationship between inputs and labels. However, the observed High-class F_2 was only around the 45th percentile, providing little evidence that the baseline reliably identified High-severity observations better than the permutation-based null models.

The XLM-RoBERTa model with child age showed much stronger separation from the null distributions. Its observed Macro-F1 of 0.518 exceeded all 100 permutation runs, whose maximum was 0.490. Likewise, its High-class F_2 of 0.510 exceeded all permutation results, whose maximum was 0.408. This provides stronger evidence that the advanced model captured non-random predictive information in the available inputs. However, this result should still be interpreted together with the train-test robustness analysis in Section 5.2.

6.2 Quantitative Error Analysis

We next examined how prediction errors were distributed across the three severity classes. Table 4 compares the confusion matrices of the logistic-regression baseline and XLM-RoBERTa with child age on the primary test set. Each cell reports both the number of observations and the percentage within the corresponding true class.

The logistic-regression baseline showed particular difficulty with the High class. It correctly classified only 2 of the 9 High observations, while 6 were

misclassified as Low. In contrast, XLM-RoBERTa with child age correctly identified 5 of the 9 High observations and misclassified only 2 as Low.

As reported in Table 2, this difference was reflected in the class-specific metrics: High-class recall increased from 0.222 to 0.556 and precision from 0.154 to 0.385. Importantly, both models produced 13 High predictions in total, indicating that the improvement was not simply due to predicting the High class more frequently, but to assigning the High label more accurately.

Overall, the quantitative error analysis shows that the main improvement of the advanced model on the primary split was its better discrimination of High-severity observations, while also correctly classifying more Low and Medium observations. Nevertheless, because the test set contained only nine High observations, these class-specific differences remain sensitive to a small number of individual predictions and should be interpreted cautiously.

6.3 Qualitative Analysis

To better understand the models' behavior beyond aggregate performance metrics, we manually examined representative correct and incorrect predictions from both the logistic-regression baseline and the XLM-RoBERTa model with child age. Comparing similar types of examples across the two models suggested several recurring error patterns.

Positivity mask. One pattern involved High-CBCL observations in which the *Event* field described severe distress or clear child difficulties, while the subsequent *EER_text* contained reassuring, hopeful, or self-supportive language. For example, one High-CBCL observation described a daughter experiencing suicidal thoughts, followed by a letter emphasizing that the parent was a good mother and would have the strength to support her child. XLM-RoBERTa correctly favored the High class in this case ($P(\text{High}) = 0.346$), correcting the baseline error. In another High-CBCL example describing fears, anxiety, restlessness, and sleep difficulties, followed by the optimistic statement that "better days will come," XLM-RoBERTa still favored Low ($P(\text{Low}) = 0.403$). These examples suggest that positive language in the emotion-regulation letter may, in some cases, partially mask severe child-related content appearing in the Event description.

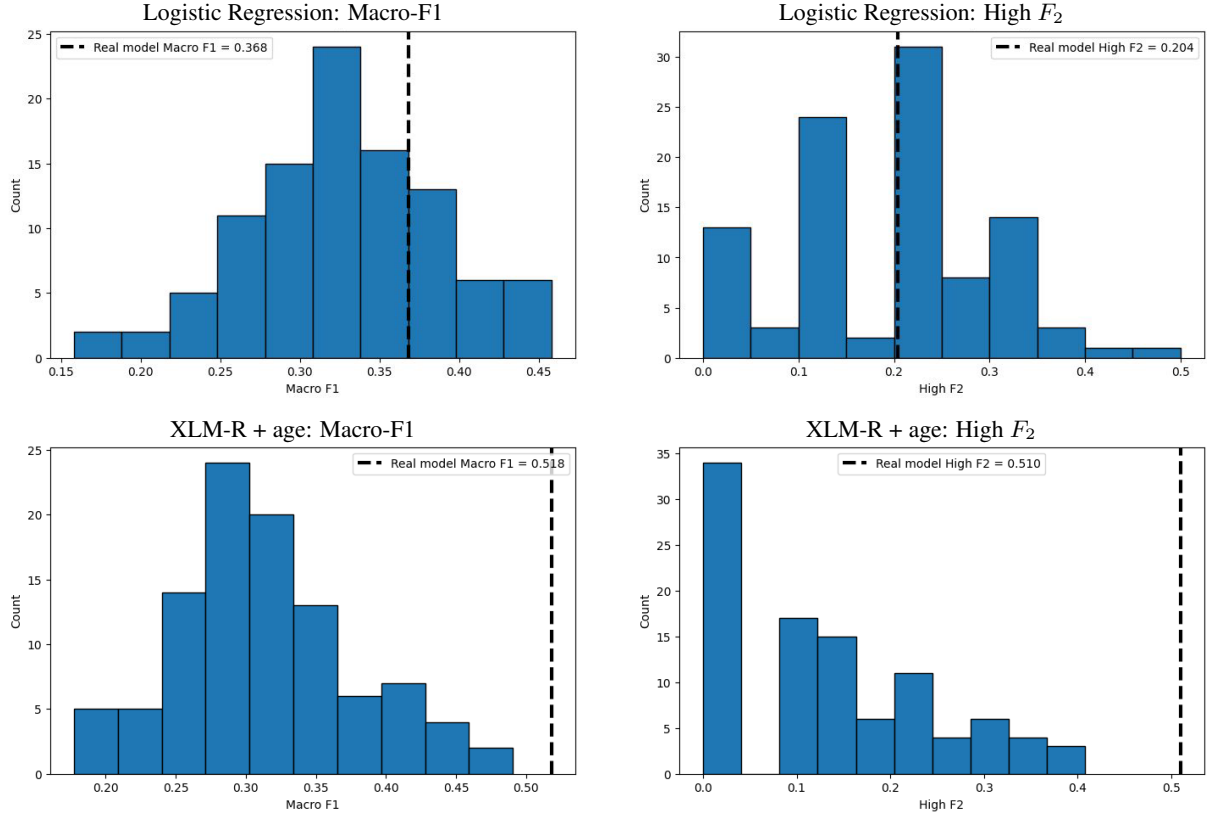


Figure 3: Permutation-based null distributions for the logistic-regression baseline and XLM-RoBERTa with child age, based on 100 random permutations of the training labels. Dashed vertical lines indicate performance obtained using the true labels. The XLM-RoBERTa model exceeded all 100 permuted-label runs for both Macro-F1 and High-class F_2 , whereas the logistic-regression baseline showed substantially weaker separation from its null distributions.

Acute war-related context. A second pattern involved low-CBCL observations containing highly salient war-related language. The baseline misclassified examples describing combat, soldiers, and intense parental fear as indicating greater child difficulties. For example, a parent describing prolonged uncertainty while a son was fighting in Gaza had a CBCL score of 11, while another observation described a son serving as a combat soldier in the Nahal Brigade and substantial parental stress despite a CBCL score of 6. XLM-RoBERTa corrected one of these illustrative errors, while the other remained misclassified. This suggests that acute war-related parental distress may sometimes be difficult to distinguish from evidence of severe child behavioral difficulties, although the advanced model appeared better able to capture this distinction in some cases.

Parental guilt and loss of control. A third pattern involved parents describing anger, impatience, or loss of control toward their children despite relatively low CBCL scores. The baseline misclassified several inspected examples of this type toward the High class. The advanced model showed mixed

behavior. In one example involving a parent becoming angry after difficulties putting a young child to sleep, XLM-RoBERTa still assigned its highest probability to High ($P(\text{High}) = 0.399$), despite a CBCL score of 13. In another example involving frustration directed toward the children, the advanced model correctly favored Low. These examples suggest that emotionally intense descriptions of parental behavior may be difficult to disentangle from the child’s actual behavioral and emotional state.

Overall, the qualitative analysis points to a possible broader challenge in the prediction task. The text fields primarily describe the parent’s experience, emotional reaction, and coping process, whereas the prediction target measures the child’s behavioral and emotional difficulties. Linguistic cues associated with parental distress may therefore sometimes increase predicted severity, while reassuring language may obscure severe child-related content. XLM-RoBERTa appeared to resolve some of the baseline errors, but the persistence of similar mistakes suggests that this underlying ambiguity

Logistic Regression – Text				
True class	Predicted class			
	Low	Medium	High	
Low	12	7	5	(50.0%)
	(50.0%)	(29.2%)	(20.8%)	
Medium	7	8	6	(33.3%)
	(33.3%)	(38.1%)	(28.6%)	
High	6	1	2	(66.7%)
	(66.7%)	(11.1%)	(22.2%)	

XLM-RoBERTa – Text + Age				
True class	Predicted class			
	Low	Medium	High	
Low	15	5	4	(62.5%)
	(62.5%)	(20.8%)	(16.7%)	
Medium	8	9	4	(38.1%)
	(38.1%)	(42.9%)	(19.0%)	
High	2	2	5	(22.2%)
	(22.2%)	(22.2%)	(55.6%)	

Table 4: Confusion matrices for the logistic-regression baseline and XLM-RoBERTa with child age on the primary held-out test set. Each cell reports the number of observations and, in parentheses, the percentage within the corresponding true class.

remained present in the advanced model.

7 Research Task: Ordinal Classification with CORN

7.1 Motivation and Method

The three target classes used in this study have a natural ordering: *Low* < *Medium* < *High*. However, the standard multiclass classifiers used in the main experiments do not explicitly use this ordering. Under a conventional multiclass objective, predicting Low instead of High is not structurally distinguished from predicting Medium instead of High, even though the former represents a larger error in severity.

To investigate whether explicitly modeling this ordinal structure could improve prediction, we applied the Conditional Ordinal Regression for Neural Networks (CORN) framework proposed by Shi et al. (2023). CORN reformulates an ordinal prediction problem with K ordered classes into $K - 1$ conditional binary tasks. Rather than estimating each ordinal boundary independently, it estimates conditional probabilities and combines them through the chain rule. This construction produces rank-consistent ordinal predictions without requiring the weight-sharing constraint used by earlier ordinal neural-network methods such as CORAL (Cao et al., 2020).

This formulation is well suited to our target because the Low, Medium, and High groups represent increasing levels of behavioral and emotional difficulty, while the distances between adjacent groups are not assumed to be equal. Moreover, because the groups were defined using training-set quantiles rather than clinical thresholds, they should be interpreted as relative severity groups rather than clinically defined categories.

7.2 Application to the CBCL Prediction Task

We implemented CORN using the same XLM-RoBERTa backbone and the same text-plus-age input representation as the strongest model in the primary classification experiment.

Because the prediction problem contains three ordered classes, CORN models two ordinal tasks corresponding to the boundaries between Low, Medium, and High. The model was trained using the same XLM-RoBERTa configuration as the primary transformer models, as detailed in Appendix A. During inference, a threshold of 0.5 was used to convert the ordinal probabilities into one of the three severity levels.

Keeping the backbone, input representation, and training configuration fixed allowed us to examine whether replacing the standard multiclass objective with an ordinal objective improved prediction.

7.3 Results

The ordinal formulation did not improve performance on our data. On the held-out test set, the CORN model predicted *Medium* for all 54 observations.

This result represents a clear failure mode rather than an improvement over the standard classification formulation. CORN correctly exploits the fact that the target classes are ordered, but in our setting, the ordinal structure was not sufficient to overcome weak or unstable predictive signal. In our training data, the High class contained relatively few unique observations, and oversampling increased its frequency only by duplicating existing examples rather than introducing new information.

A second possible explanation concerns the relationship between the textual input and the prediction target. As discussed in Section 6.3, the parental texts frequently contain strong signals of parental distress, coping, and war-related context that do not map directly onto the child’s CBCL

severity. An ordinal objective can exploit the ordering among severity levels, but it cannot create a sufficiently informative relationship between the predictors and the outcome when that relationship is weak or ambiguous in the available data.

Overall, the CORN formulation did not provide a useful separation of the three severity groups in this dataset. This negative result illustrates that incorporating theoretically appropriate target structure does not necessarily improve predictive performance, particularly when the available sample is small, the rarest class contains few unique observations, and the connection between the predictors and the target is indirect.

8 Conclusions and Limitations

This study examined whether parent-authored free text collected during wartime could be used to predict children’s behavioral and emotional difficulties as measured by the CBCL. Overall, the results suggest that the parental texts contained meaningful predictive information, but that the relationship between the textual input and the child-level outcome was complex and not consistently captured across models and data splits.

Among the primary model configurations, XLM-RoBERTa with child age achieved the strongest performance on the original 80/20 split, including the best Macro-F1 and High-class F_2 . Its observed performance also exceeded all 100 label-permutation runs for both metrics, providing evidence of non-random predictive signal. However, the benefit associated with child age was not stable across alternative train–test splits. The primary result should therefore be interpreted as evidence of useful predictive signal rather than as a robust and general advantage of adding child age.

The qualitative analysis suggested several possible sources of model error. In some inspected cases, reassuring language appeared to obscure severe child-related content, while highly salient war-related language and parental distress sometimes coincided with overestimation of child difficulty. XLM-RoBERTa appeared to resolve some of these baseline errors, but the broader ambiguity between parental emotional distress and the child’s behavioral state remained.

The ordinal structure of the target did not by itself resolve the prediction difficulty. Although CORN was conceptually well matched to the ordered Low–Medium–High outcome, the model col-

lapsed to predicting the Medium category for every test observation, indicating that explicitly modeling class ordering was not sufficient in this setting.

Several limitations constrain the interpretation of the results. First, the final dataset contained only 270 observations, with relatively few High-severity cases. As a result, both training and evaluation of the High class were sensitive to individual observations. Oversampling reduced class imbalance during training, but did not add new information. Second, the target classes were defined using quantiles of the training CBCL distribution rather than clinically validated thresholds. The Low, Medium, and High labels should therefore be interpreted as project-specific relative severity groups rather than clinical categories.

Third, the available text provides only an indirect view of the prediction target. Much of the input reflects the parent’s experience, distress, and coping, whereas the CBCL measures the child’s behavioral and emotional difficulties. This mismatch may limit predictability even when the text contains meaningful signal.

One important lesson from this project was working, for the first time, with a very small dataset. This made the trade-off between training and evaluation especially clear: allocating more observations to training gives the model more examples to learn from, while a larger test set provides a more reliable basis for evaluating performance. The project therefore highlighted how strongly data allocation can influence conclusions when the available sample is limited.

More broadly, the project highlighted the importance of evaluating models beyond a single aggregate score. The main conclusions depended not only on Macro-F1, but also on High-class recall and F_2 , robustness across alternative data splits, and qualitative examination of model errors. These analyses revealed limitations that would have been obscured by reporting only the best-performing configuration. In a small and imbalanced dataset, examining when and why a model fails was therefore essential for interpreting apparent performance gains.

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A XLM-RoBERTa Experimental Setup

Table 5 summarizes the configuration used in the primary XLM-RoBERTa experiments.

Setting	Value
Maximum sequence length	384
Training epochs	20
Batch size	8
Learning rate	10^{-3}

Table 5: Primary XLM-RoBERTa training configuration.

The structured input format was:

EVENT: <event text>

EER_TEXT: <emotion-regulation text>

METADATA: child_age=<age>

The metadata line was omitted in the text-only configuration. All other training settings were kept identical between the two feature conditions.